

Daniel Smullen

Curriculum Vitae

"Don't have good ideas if you aren't willing to be responsible for them." —Alan Perlis

About Me

I design privacy-enhancing technologies, interoperable standards, usable privacy and security controls, and AI-enabled trust systems for connected-device ecosystems, smart homes, and real-world infrastructure. My work combines software engineering, privacy engineering, human-computer interaction, requirements engineering, threat modeling, machine learning, data governance, Internet of Things (IoT) systems, and technical standards.

I help organizations develop usable, secure, privacy-preserving, and trustworthy software systems.

Education

- PhD **Software Engineering**, *Carnegie Mellon University*, Pittsburgh
Software and Societal Systems Department (formerly Institute for Software Research), Committee: Norman Sadeh (Chair), Lorrie Faith Cranor, Alessandro Acquisti, Rebecca Weiss (External, Mozilla), Yaxing Yao (External, UMBC)
○ Research focused on usable privacy and security across browsers, mobile apps, and the Internet of Things, using mixed methods from HCI, behavioral economics, user-centered design, requirements engineering, machine learning, and empirical software engineering.
○ Dissertation work assessed privacy and security controls for effectiveness, manageability, public policy implications, and alignment with people's mental models; it also showed how machine learning can reduce user burden while preserving preference accuracy.
- MSc **Software Engineering**, *Carnegie Mellon University*, Pittsburgh, Software and Societal Systems Department
- BEng **Software Engineering, Honours with Distinction**, *Ontario Tech University (formerly University of Ontario Institute of Technology)*, Oshawa

Professional Work Experience

- 2024–Present **Principal Privacy Research Engineer**, *CableLabs, Security & Privacy Technologies*, Remote
Lead privacy and trust R&D for connected-device ecosystems, IoT, smart homes, privacy-enhancing technologies, usable privacy and security, and technical standards.
○ Report to CableLabs CISO Brian Scriber and translate privacy goals into architectures, protocol requirements, data taxonomies, AI-aware usability guidance, and engineering controls.
○ Represent CableLabs in IETF Internet-Drafts and Connectivity Standards Alliance standards work across Matter, Zigbee, Aliro, and Data Privacy Working Group activity.
○ Chair the Connectivity Standards Alliance Marketing and Product Subgroup and Data Export Strategic Incubator Tiger Team; advise partners, executives, product strategy, public policy, and standards participants.

- 2023–Present **Founding Ambassador and Standards Committee Member**, *Institute of Operational Privacy Design (IOPD)*, Volunteer
 Volunteer Founding Ambassador and Standards Committee Member developing privacy design standards that make privacy-by-design and privacy-by-default claims auditable and implementable.
 ○ Translate privacy principles, controls, evidence requirements, assurance criteria, and certification expectations into practical standards language.
- 2023–2024 **Applied Scientist II**, *Amazon Lab126, Alexa Sensitive Content Intelligence*, Remote
 Built AI-powered and LLM-based systems for large-scale Alexa customer feedback analysis, including an Alexa+ pipeline that framed chat transcripts as reading-comprehension and information-extraction tasks.
 ○ Used off-the-shelf Large Language Models, prompt engineering, and structured extraction to identify trust, safety, and product-quality signals from conversational feedback.
 ○ Replaced custom BERT-style model development and training-data pipelines, reducing data labeling, engineering cost, and maintenance burden.
 ○ Modeled customer-trust risk in Alexa interactions using privacy metrics, prompt design, and mixed-methods research.
- 2021–2023 **Applied Scientist II**, *Amazon Lab126, Devices & Services Trust and Privacy*, Remote
 Worked across Amazon Devices & Services on technical privacy and trust problems spanning research, privacy engineering, legal decision-making, and product execution.
 ○ Built interpretable machine learning and decision-tree models connecting data type classification, legal attestation decisions, PII prediction, and model explanations.
 ○ Served as penholder for an internal generative AI data use policy, helping translate privacy, legal, data governance, and product-risk requirements into implementable guidance for AI-enabled product teams.
 ○ Designed privacy-preserving architectures, Privacy Enhancing Technologies (PETs), compliance tooling, privacy metrics, and privacy requirements engineering guidance; served as a privacy engineering subject matter expert for product, legal, leadership, and engineering teams on Privacy by Design, generative AI, and risk.
- 2015 **334F Research Associate (Radar Science and Instrument Engineering)**, *NASA Jet Propulsion Laboratory*, Pasadena
 Supervisor: Razi Ahmed
 ○ Developed high-performance, resolution-preserving InSAR image processing for UAVSAR radar imagery.
 ○ Implemented cross-platform Python/C11 software with OpenMP, MPI, and NASA Ames Pleiades high-performance computing support.
 ○ Reduced practical processing time from weeks to minutes and contributed production radar science software.
- 2012 **Engineer-in-Training (Co-op)**, *SNC-Lavalin, Global Information Technologies*, Toronto
 ○ Assessed mission-critical datacenter reliability, disaster recovery, emergency response, infrastructure risk, and Splunk-based operational performance.
 ○ Recommended infrastructure improvements, risk mitigations, workstation staging automation, and legacy DOS system replacements.
 ○ Reduced workstation deployment time from hours to minutes.
- 2011 **Support Technician**, *SNC-Lavalin, Mining and Metallurgy*, Toronto
 ○ Provided dedicated technical support for high-value clients, engineering departments, and senior management.
 ○ Developed automated data recovery workflows and performed high-risk computer forensics on damaged hardware.
 ○ Delivered internal seminars on computer and network security, system repair, open source software, and computer forensics.

- 2009 **Laboratory Systems Administrator**, *DESSAU (LVM-Technisol)*, Toronto
- Managed migration and replacement of laboratory information management systems during the JEGEL/LVM-Technisol and DESSAU integration.
 - Supported geotechnical field operations, subsurface sampling, concrete and asphalt QA/QC testing, and airport and highway resurfacing projects.
 - Bridged laboratory information management, field data, and IT systems during active civil infrastructure work.
- 2009 **Laboratory Systems Administrator**, *John Emery Geotechnical Engineering Ltd.*, Toronto
- Supported City of Toronto special projects, the Toronto 2009 Capital Works Program, and Trans-Canada Highway Rehabilitation work.
 - Managed geoinformatics, utility-location coordination, subsurface-sampling safety workflows, and collaborative development of ISO 9001 certified laboratory sample management software.
 - Repaired legacy pavement-analysis embedded systems and designed file-storage, database, and datacenter infrastructure for the Toronto engineering office.

Academic Work Experience

- 2021 **Postdoctoral Researcher**, *Carnegie Mellon University*, Pittsburgh, Institute for Software Research / S3D (The Software and Societal Systems Department)
- Researched user-centered browser and web-platform privacy and security controls, extending dissertation work in usable privacy and security, HCI, privacy engineering, and control design.
- 2014–2021 **Graduate Research Assistant**, *Carnegie Mellon University*, Pittsburgh, Institute for Software Research / S3D (The Software and Societal Systems Department)
- Conducted Ph.D. research in usable privacy and security using mixed methods from HCI, behavioral economics, user-centered design, requirements engineering, machine learning, and empirical software engineering. Produced publications, datasets, and prototype systems that informed policy discussions, privacy-engineering practice, and usable privacy and security standards work.
- 2017–2021 **Research Advisor**, *Carnegie Mellon University*, Pittsburgh, Institute for Software Research / S3D (The Software and Societal Systems Department)
- Mentored undergraduate research programmers and M.Sc. students.
- Trained researchers in Agile, pair programming, extreme programming, test-driven development, debugging, and version control.
 - Provided hands-on research methods training in crowdsourcing, survey design, contextual interviews, and grounded/thematic analysis.
- 2016–2020 **Teaching Assistant**, *Carnegie Mellon University*, Pittsburgh, Institute for Software Research / S3D (The Software and Societal Systems Department)
- 17-655/17-755: Architectures for Software Systems
 - 17-781/45-887: Mobile and IoT Computing Services
- 2013–2014 **Research Associate**, *UOIT (now Ontario Tech University)*, Oshawa
- Software Quality Research Lab, Advisor: Jeremy Bradbury
- Investigated privacy exposure in network metadata released through ordinary Internet traffic in an ethics board-approved user study.
 - Studied the software testing coupling effect using mutation testing, automated software analysis, and software-quality research methods.
 - Research funded by NSERC/CRSNG (Canada).
- 2013 **Undergraduate Research Assistant**, *UOIT (now Ontario Tech University)*, Oshawa
- Software Engineering Lab, Advisor: Ramiro Liscano
- Built TinyOS-based wireless sensor network tooling for policy programming, including Policy IDE, TOSServ, and IPv6-enabled Finger2IPv6 remote development variants for embedded systems.
 - Explored how policy-based programming could make distributed embedded systems easier to develop, debug, and manage.
 - Research funded by NSERC/CRSNG (Canada) Undergraduate Student Research Award.

Projects and Funded Research

- 2025–Present **Privacy Preference Declaration (PPD) for Home Networks**, *CableLabs*, Remote, Co-author: Brian Scriber
Standards and privacy engineering project for user-defined privacy preferences in home networks, including IETF Internet-Drafts on protocol architecture, device accountability, signed receipts, and a machine-readable taxonomy for device data handling practices.
- 2025 **Defining Privacy Engineering as a Profession**, *Carnegie Mellon University*, Pittsburgh
Qualitative privacy engineering study identifying practitioner responsibilities, collaboration patterns, core competencies, hiring expectations, and professionalization challenges. Funded by: NSF CCF-2217771.
- 2023–2024 **Alexa+ Feedback Processing Pipeline**, *Amazon Lab126*, Remote
Internal applied machine learning project using off-the-shelf LLMs, prompt engineering, reading-comprehension framing, and structured extraction to process Alexa+ chat transcripts and customer feedback without custom BERT-style models or dedicated training datasets.
- 2022–2023 **Privacy Attestation PII Prediction and Explanation**, *Amazon Lab126*, Remote
Internal privacy engineering project using data type classification, legal attestation decisions, interpretable decision trees, PII prediction, and model explanations to turn privacy review patterns into actionable engineering signals.
- 2023 **Generative AI Data Use Policy**, *Amazon Lab126*, Remote
Internal privacy policy and data governance initiative where I served as penholder for generative AI data use guidance, helping translate privacy, legal, responsible AI, customer-trust, and product-risk requirements into implementable guidance for AI-enabled product teams.
- 2020–2022 **Usable Privacy and Security Controls**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Research thread on browser privacy controls, web privacy tools, Tor Browser adoption, mobile-payment security nudges, and user-centered privacy and security control design. Funded by: DARPA/AFRL FA8750-15-2-0277; NSF CNS-15-13957, CNS-1801316, CNS-1914486, CNS-1330596, and SES-1513957.
- 2020–2021 **Engineering Usable Privacy and Security Affordances for Notice and Choice**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
User-centered privacy and security affordances for notice, choice, preference management, and control interfaces. Funded by: DARPA/AFRL FA8750-15-2-0277; NSF CNS-15-13957, CNS-1801316, CNS-1914486.
- 2019–2020 **Design and Evaluation of Security and Privacy Nudges**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Informational, planning, privacy, and security nudges that help users translate privacy and security intentions into action. Funded by: NSF CNS-1330596, SES-1513957, CNS-1801316.
- 2019 **Mobile App Privacy System (MAPS)**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Large-scale mobile app privacy compliance analysis system for the Google Play Store ecosystem, connecting crawl infrastructure, privacy policies, code analysis, and enforcement support. Funded by: NSF CNS-1330596, CNS-1330141, CNS-1330214; XSEDE ACI-1548562.
- 2017–2019 **IoT Resource Registry**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Personalized Privacy Assistant and IoT Privacy Infrastructure project for machine-readable service descriptions, contextual notice, choice, and privacy preference management in connected environments. Funded by: DARPA/AFRL FA8750-15-2-0277; NSF CNS-15-13957, CNS-1801316, CNS-1914486.

- 2018–2019 **Internet of Things Privacy Infrastructure (Patent Pending)**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
IoT privacy infrastructure, resource registries, service descriptions, and mechanisms for contextual notice and choice. Funded by: DARPA/AFRL FA8750-15-2-0277; NSF CNS-15-13957, CNS-1801316, CNS-1914486.
- 2017–2019 **DARPA Brandeis: A Privacy Assistant for the Internet of Things**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Privacy assistant infrastructure for nearby IoT services, sensors, smart spaces, and context-aware systems. Funded by: DARPA/AFRL FA8750-15-2-0277; NSF CNS-15-13957, CNS-1801316, CNS-1914486.
- 2017–2019 **The Usable Privacy Policy Project**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
Privacy policy analysis project combining crowdsourcing, natural language processing, machine learning, privacy policy annotations, mobile app privacy compliance, and usable privacy policy research. Funded by: NSF CNS-1330596, CNS-1330141, CNS-1330214.
- 2017 **Mobile App Privacy Compliance System (Patent Pending)**, *Carnegie Mellon University*, Pittsburgh, Advisor: Norman Sadeh
MAPS precursor for scalable mobile app privacy compliance analysis across code-level practices and privacy policy disclosures. Funded by: NSF CNS-1330596, CNS-1330141, CNS-1330214; XSEDE ACI-1548562.
- 2016 **Eddy: A Privacy Requirements Specification Language**, *Carnegie Mellon University*, Pittsburgh, Advisor: Travis Breaux
Privacy requirements engineering language and formal analysis tool for collection, use, transfer, retention, compliance reasoning, and consistency checking over privacy obligations. Funded by: NSF CNS-1330596; ONR N002441410028; National Security Agency.
- 2015 **Non-Local Interferometric SAR Parameter Estimator**, *NASA Jet Propulsion Laboratory*, Pasadena, Advisor: Razi Ahmed
High-performance, resolution-preserving InSAR filtering system for UAVSAR imagery, using cross-platform Python/C11, OpenMP, MPI, and supercomputing deployment to reduce processing time from weeks to minutes. Funded by: NASA JPL PhD Fellowship.
- 2015 **NL-InSAR with Gaussian-Laplacian Pyramids**, *NASA Jet Propulsion Laboratory*, Pasadena, Advisor: Razi Ahmed
Radar image processing project extending non-local InSAR filtering with Gaussian-Laplacian pyramids to reduce canvas effects while preserving reflectivity, interferometric phase, coherence, and fine image structure.
- 2014 **Incident Recognition and Intelligence System (IRIS)**, *Ontario Tech University*, Oshawa, Advisor: Shahryar Rahnamayan
Computer vision, web, and traffic-monitoring system for highway incident recognition, video review, statistical aggregation, and situational awareness. Funded by: University of Toronto ITS Laboratory, Ontario Tech University, NSERC/CRSNG.
- 2014 **Automated Marking System (AMS)**, *Ontario Tech University*, Oshawa, Advisor: Kamran Sartipi
Ruby on Rails educational technology system for creating, distributing, compiling, executing, and automatically grading introductory programming assignments in a protected environment. Funded by: Ontario Tech University.
- 2013 **Policy IDE, TOSServ, Finger2IPv6**, *Ontario Tech University*, Oshawa, Advisor: Ramiro Liscano
TinyOS wireless sensor network tooling for policy-based programming, debugging policy-driven embedded applications, server-side TinyOS workflows, and IPv6-enabled remote development. Funded by: NSERC/CRSNG Undergraduate Student Research Award.

- 2013 **Military Logistics Management System (MLMS)**, *Ontario Tech University*, Oshawa, Advisor: Eyhab Al-Masri
ASP.NET military logistics dashboard and visualization system for supply, personnel, operational reporting, role-aware access, and command-and-control style situational awareness. Funded by: Ontario Tech University.
- 2012 **sneakyFS File System Design and Implementation**, *Ontario Tech University*, Oshawa, Advisor: Kamran Sartipi
UNIX-style C file system on a simulated hard disk with indexed and linked allocation, transparent encryption, journaling, UUIDs, low-level memory management, and secure storage design. Funded by: Ontario Tech University.
- 2012 **Datacenter Utilization Research Study**, *SNC-Lavalin Global Information Technologies*, Toronto, Advisor: Marc Ross
Mission-critical datacenter utilization, availability, performance, disaster recovery, and infrastructure remediation study for critical server assets at the West Mall campus datacenter. Funded by: SNC-Lavalin.
- 2009 **Laboratory Information Management System (LIMS)**, *John Emery Geotechnical Engineering Ltd.*, Toronto, Advisor: John Emery
ISO 9000-certified laboratory information management software for geotechnical and materials-testing workflows, including sample intake, process queues, laboratory status tracking, and production reporting.
- 2009 **Trans-Canada Highway Rehabilitation Geoinformatics**, *John Emery Geotechnical Engineering Ltd.*, Toronto, Advisor: Alain Duclos
Geoinformatics and civil engineering support project for large-scale asphalt rehabilitation planning across more than 300 km of the Trans-Canada Highway near Banff, including technical drawings, probe-hole data, pavement conditions, and remediation planning.

Refereed Publications

- D. Smullen**, "Privacy Preference Declarations: Signed receipts for expectation-aligned smart homes," in *2025 IEEE Symposium on Privacy Expectations (ISoPE)*. IEEE, 2025, pp. 14–17.
- J. M. del Alamo, S. Human, K. Kollnig, **D. Smullen**, and K. Vaniea, "Cross-stakeholder interaction," in *Grand Challenges for Research on Privacy Documents (Dagstuhl Seminar 25021)*, ser. Dagstuhl Reports, F. Schaub, C. Utz, S. Wilson, and L. Xian, Eds. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025, vol. 15, no. 1, pp. 29–32.
- N. Samarin, N. R. Narla, L. Webster, and **D. Smullen**, "Defining privacy engineering as a profession," *Proceedings on Privacy Enhancing Technologies*, vol. 2025, no. 4, pp. 549–565, 2025.
- P. Story, **D. Smullen**, R. Chen, Y. Yao, A. Acquisti, L. F. Cranor, N. Sadeh, and F. Schaub, "Increasing adoption of Tor Browser using informational and planning nudges," *Proceedings on Privacy Enhancing Technologies*, vol. 2022, no. 2, pp. 152–183, 2022.
- D. Smullen**, "Informing the design and refinement of privacy and security controls," PhD dissertation, Carnegie Mellon University, Sep. 2021.
- P. Story, **D. Smullen**, Y. Yao, A. Acquisti, L. F. Cranor, N. Sadeh, and F. Schaub, "Awareness, adoption, and misconceptions of web privacy tools," *Proceedings on Privacy Enhancing Technologies*, vol. 2021, no. 3, pp. 308–333, 2021.
- D. Smullen**, Y. Yao, Y. Feng, N. Sadeh, A. Edelstein, and R. Weiss, "Managing potentially intrusive practices in the browser: A user-centered perspective," *Proceedings on Privacy Enhancing Technologies*, vol. 2021, no. 4, pp. 500–527, 2021.
- P. Story, **D. Smullen**, A. Acquisti, L. F. Cranor, N. Sadeh, and F. Schaub, "From intent to action: Nudging users towards secure mobile payments," in *Sixteenth Symposium on Usable Privacy and Security (SOUPS 2020)*, 2020,

pp. 379–415. [Online]. Available: <https://www.usenix.org/conference/soups2020/presentation/story>

D. Smullen, Y. Feng, S. A. Zhang, and N. Sadeh, “The best of both worlds: Mitigating trade-offs between accuracy and user burden in capturing mobile app privacy preferences,” *Proceedings on Privacy Enhancing Technologies*, vol. 2020, no. 1, pp. 195–215, 2020.

S. Zimmeck, P. Story, **D. Smullen**, A. Ravichander, Z. Wang, J. Reidenberg, N. C. Russell, and N. Sadeh, “MAPS: Scaling privacy compliance analysis to a million apps,” *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 3, pp. 66–86, 2019.

P. Story, S. Zimmeck, A. Ravichander, **D. Smullen**, Z. Wang, J. Reidenberg, N. C. Russell, and N. Sadeh, “Natural language processing for mobile app privacy compliance,” in *AAAI Spring Symposium on Privacy-Enhancing Artificial Intelligence and Language Technologies*, 2019. [Online]. Available: https://ceur-ws.org/Vol-2335/1st_PAL_paper_6.pdf

S. Wilson, F. Schaub, F. Liu, K. M. Sathyendra, **D. Smullen**, S. Zimmeck, R. Ramanath, P. Story, F. Liu, N. Sadeh, and N. A. Smith, “Analyzing privacy policies at scale: From crowdsourcing to automated annotations,” *ACM Transactions on the Web*, vol. 13, no. 1, Dec. 2018.

A. Das, M. Degeling, **D. Smullen**, and N. Sadeh, “Personalized privacy assistants for the internet of things: Providing users with notice and choice,” *IEEE Pervasive Computing*, vol. 17, no. 3, pp. 35–46, 2018.

D. Smullen and T. Breaux, “Improving security in software acquisition with data retention specifications,” US Navy Acquisition Research Program Sponsored Report Series, Tech. Rep., 2017. [Online]. Available: <https://calhoun.nps.edu/handle/10945/58892>

J. Bhatia, T. Breaux, L. Friedberg, H. Hibshi, and **D. Smullen**, “Privacy risk in cybersecurity data sharing,” in *ACM Workshop on Information Sharing and Collaborative Security (WISCS)*, 2016, pp. 57–64.

D. Smullen and T. Breaux, “Modeling, analyzing, and consistency checking privacy requirements using Eddy,” in *Proceedings of the Symposium and Bootcamp on the Science of Security*, 2016, pp. 118–120.

T. Breaux, **D. Smullen**, and H. Hibshi, “Detecting repurposing and over-collection in multi-party privacy requirements specifications,” in *IEEE International Requirements Engineering Conference*, 2015, pp. 166–175.

D. Smullen and T. Breaux, “Toward rapid recertification using formal analysis,” US Navy Acquisition Research Program Sponsored Report Series, Tech. Rep., 2015. [Online]. Available: <https://www.acquisitionresearch.net/publications/detail/1532/>

D. Smullen, J. Gillett, J. Heron, and S. Rahnamayan, “Genetic algorithm with self-adaptive mutation controlled by chromosome similarity,” in *IEEE Congress on Evolutionary Computation (CEC)*, 2014, pp. 504–511.

N. Qwasmī, **D. Smullen**, and R. Liscano, “Integrated development environment for debugging policy-based applications in wireless sensor networks,” *Procedia Computer Science*, vol. 21, pp. 225–233, 2013.

Standards and Technical Drafts

Privacy Preference Declaration for Home Networks, *IETF Internet-Draft, work in progress*. Authored with CableLabs CISO Brian Scriber. Defines a framework for user-defined privacy preferences across home networks, including policy discovery, device accountability, and privacy governance for connected-device ecosystems.

Privacy Preference Declaration Taxonomy, *IETF Internet-Draft, work in progress*. Authored with CableLabs CISO Brian Scriber. Defines a machine-readable taxonomy for describing device data handling practices, including data types, purposes, actions, sources, and destinations.

Talks, Posters, Workshops, and Recorded Sessions

- 2026 **PETs in Practice: From Privacy Policies to User-Held Preferences with CableLabs, Duality Technologies PETs in Practice**, Online, Recorded Series
Privacy-enhancing technologies, user-held preferences, and privacy engineering deployment for connected-device ecosystems.
- 2025 **AI's Best Friend: The Role of PETs, Enabling Data Use & Operationalizing Privacy, IAPP Privacy. Security. Risk. 2025**, San Diego, Panel
Expert panel on Privacy Enhancing Technologies, privacy operations, responsible data use, and privacy engineering practice.
- 2025 **Grand Challenges for Research on Privacy Documents, Dagstuhl Seminar 25021**, Wadern, Germany, Invited Research Seminar
Invited participant in interdisciplinary research-roadmap work on privacy policies, terms of use, cookie notices, NLP, machine learning, HCI, public policy, usable notice and consent, standardization, datasets, automation, and cross-stakeholder engagement.
- 2025 **Privacy Preference Declarations: Signed Receipts for Expectation-Aligned Smart Homes, IEEE Symposium on Privacy Expectations (ISoPE 2025)**, New York City, Paper Session
Presented peer-reviewed paper on Privacy Preference Declarations, signed receipts, smart-home expectations, and user-centered privacy governance for connected devices.
- 2023 **Data-Driven Methodologies Toward More Usable Personalized Settings, Amazon DataCon**, Online, Conference Talk
Presented award-winning work on usable privacy and security, machine learning, and mixed-methods research.
- 2022 **Information Privacy Engineering, University of Maine School of Computing and Information Science**, Online, Guest Lecture
Graduate lecture and discussion on information privacy engineering.
- 2020 **The Best of Both Worlds: Mitigating Trade-offs Between Accuracy and User Burden in Capturing Mobile App Privacy Preferences, 20th Privacy Enhancing Technologies Symposium (PETS 2020)**, Online, Conference Talk
Presented peer-reviewed paper on mobile app privacy preferences, machine learning, accuracy, and user burden.
- 2019 **Digital Data Flows Masterclass: Mobile Apps, Future of Privacy Forum**, Online, Masterclass
Masterclass on mobile app data flows, privacy compliance, and usable privacy.
- 2018 **Personalized Privacy Assistant Project, US Federal Trade Commission PrivacyCon 2018**, Washington D.C., Research Poster
Poster and oral presentation on personalized privacy assistants.
- 2017 **Assisting Users in a World Full of Cameras: A Privacy-aware Infrastructure for Computer Vision Applications, US Federal Trade Commission PrivacyCon 2017**, Washington D.C., Research Poster
Poster and oral presentation on privacy-aware infrastructure for computer vision and IoT cameras.
- 2017 **A Privacy Assistant for the Internet of Things, 13th Symposium On Usable Privacy and Security (SOUPS 2017)**, San Jose, Research Poster
Distinguished poster presentation on privacy assistants for IoT services and smart spaces.
- 2016 **Toward a Semantics for Data Retention in Eddy, National Institute of Standards and Technology (NIST)**, Gaithersburg, Invited Talk
Invited research talk hosted by the NIST Applied Cybersecurity Division.

- 2015 **Privacy Engineering Tool Clinic**, *Computing Community Consortium Catalyst, Privacy by Design Workshop*, Pittsburgh, Invited Talk and Tool Demonstration
Invited talk, tool demonstration, and panel discussion on privacy engineering tools.
- 2015 **Detecting Repurposing and Over-Collection in Multi-party Privacy Requirements Specifications**, *23rd IEEE International Requirements Engineering Conference (RE 2015)*, Ottawa, Conference Talk
Presented peer-reviewed paper on privacy requirements engineering, repurposing, and over-collection.
- 2015 **Towards Rapid Re-Certification Using Formal Analysis**, *US Navy Acquisition Research Symposium*, Monterey, Conference Talk
Presented peer-reviewed work on formal analysis and software acquisition recertification.
- 2014 **Genetic Algorithm with Self-Adaptive Mutation Controlled by Chromosome Similarity**, *IEEE World Congress on Computational Intelligence (WCCI 2014)*, *Evolutionary Computation Conference (CEC 2014)*, Beijing, Conference Talk
Presented peer-reviewed paper on genetic algorithms and self-adaptive mutation.
- 2013 **How Much Do We Reveal Through Metadata? An Assessment of Online Privacy**, *IBM Consortium for Software Engineering Research (CSER 2013)*, Toronto, Preliminary Study Poster
Poster and oral presentation on network metadata privacy exposure.
- 2013 **Policy IDE... and Lessons Learned Since**, *The 4th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN-2013)*, Niagara Falls, Conference Talk
Presented peer-reviewed paper on Policy IDE and policy-based wireless sensor network programming.
- 2013 **Facilitating the Internet of Things with Policy Programming**, *UOIT (now Ontario Tech University) Undergraduate Research Showcase 2013*, Oshawa, Research Poster
Poster and oral presentation on policy programming for IoT and wireless sensor networks.

Professional Service

- 2026-Present **Program Committee Member**, *IEEE Symposium on Privacy Expectations (SoPE)*, Conference
Review submissions and contribute to program committee discussions on privacy expectations.
- 2020-Present **External Program Committee Member**, *Conference on Human Factors in Computing Systems (ACM SIGCHI)*, Conference
Review and shepherd main proceedings and Late Breaking Works submissions.
- 2021-Present **Reviewer**, *IEEE Access*, Online, Journal
Review submissions and participate in editorial discussions.
- 2021-2022 **Reviewer**, *ACM Transactions on Privacy and Security (TOPS, formerly known as TISSEC)*, Online, Journal
Reviewed submissions and participated in editorial discussions.
- 2021 **Program Chair**, *ACM SIGBOVIK*, Online, Conference
Organized conference and webcast; moderated Q&A.
- 2020-2021 **Council Member**, *Cylab Justice Equity Diversity and Inclusion (JEDI) Council*, Carnegie Mellon University
Helped establish committee work on justice, equity, diversity, and inclusion in faculty candidates, talks, hiring, and student affairs.

- 2019-2020 **Committee Member**, *Institute for Software Research Re-Branding Committee*, Carnegie Mellon University
Coordinated department re-branding research, messaging, communications, and recruiting strategy for the Software and Societal Systems Department.
- 2018-2021 **Co-Founder**, *Software and Societal Systems Department Forge Makerspace*, Carnegie Mellon University
Led makerspace layout, development, maintenance, safety training, CNC machining, 3D printing, and CAD instruction.
- 2018-2020 **Co-Founder**, *Software and Societal Systems Department Lunch and Learn Seminar Series*, Carnegie Mellon University, Seminar
Managed and taught weekly seminars on software engineering practices for researchers.
- 2018 **Program Committee Member**, *European Conference on Information Systems (ECIS 2018)*, Portsmouth, Conference
Reviewed submissions and participated in program committee discussions and shepherding.
- 2016 **Program Committee Member**, *International Workshop on Privacy Engineering (IWPE 2016)*, San Jose
Reviewed submissions and participated in privacy engineering program committee discussions and shepherding.
- 2015 **Web Chair**, *International Conference on Multicore Software Engineering, Performance, and Tools (MUSEPAT 2014)*, Hong Kong, Conference
Organized conference website and coordinated program committee operations.
- 2015 **Review Committee Member**, *Software and Societal Systems Department PhD Applicant Review Committee*, Carnegie Mellon University
Reviewed doctoral applications and participated in final admissions decisions.

Awards

- 2024 **Outstanding Contributor Award**, *Connectivity Standards Alliance Data Privacy Working Group*
- 2017 **Distinguished Poster Award**, *13th Symposium On Usable Privacy and Security (SOUPS 2017)*, San Jose
- 2017 **Hima and Jive Fellowship in Computer Science for International Students**, *Carnegie Mellon University*, Pittsburgh, Pennsylvania
- 2015 **Ready-Set-Transfer Award**, *23rd IEEE International Requirements Engineering Conference*, Ottawa, "Eddy: A privacy requirements specification language"
First place in competitive industrial panel talks.
- 2014 **Faculty of Engineering and Applied Science Undergraduate Capstone Design Challenge Winner**, *Ontario Tech University*, Oshawa, "Incident Recognition and Intelligence System (IRIS)"
First place in capstone research competition evaluated by academic and industrial experts.
- 2012–2014 **President's Honours List**, *Ontario Tech University*, Oshawa
Exceptional academic achievement, GPA above 3.7.
- 2013 **National Science and Engineering Research Council (NSERC/CRSNG) Undergraduate Research Award**, *Ontario Tech University*, Oshawa
Federally funded competitive research award supporting undergraduate engineering research.
- 2011 **Dean's Honours List**, *Ontario Tech University*, Oshawa
Exceptional academic achievement, GPA above 3.5.

- 2008 **Engineers Without Borders Design Challenge Winner**, *McMaster University*, Hamilton
First place in software requirements specification competition for AIDS medication distribution.
- 2007 **DaVinci Engineering Design Challenge Winner**, *University of Toronto*, Toronto
First place in electrical engineering and fluid dynamics design challenge for remote-controlled aquatic vehicles.

Languages and Technologies

- English **Native Proficiency**
- French **Verbal, Written Proficiency**
- German **Verbal, Written Proficiency**
- Python **Preferred Language**, *Django, Django REST Framework, scikit-learn, Cython, multiprocessing, multithreading, data analysis, machine learning, performance tuning*
- Docker **Preferred Technology**, *Containerization, DevOps workflows, dependency management, reproducible research prototypes, and cross-platform deployment*
- R **Preferred Language**, *Statistical analysis, regression modeling, data mining, exploratory analysis, and visualization*
- JavaScript **Secondary Preference**, *Full-stack web applications, MongoDB, Express, Angular, Node, geospatial mapping, and Ionic hybrid mobile apps*
- C++ **Secondary Preference**, *High-performance computing, optimized image-processing kernels, computer vision, and performance-driven systems*
- Java **Secondary Preference**, *Parsers, lexers, compilers, domain-specific languages, Eddy, OWL-DL, and Java-based research tools*

Professional Memberships

- Since 2024 **Connectivity Standards Alliance**, *Contributing Member*
Data Privacy Working Group; Matter, Zigbee, and Aliro Working Groups; Data Export Strategic Incubator Tiger Team Chair; Marketing and Product Subgroup Chair; Certification and Technical Subgroups.
- Since 2023 **Institute of Operational Privacy Design (IOPD)**, *Founding Ambassador, Standards Committee Member*
- Since 2023 **Association for Computing Machinery (ACM)**, *Professional Member*
- Since 2021 **International Association of Privacy Professionals (IAPP)**, *Member*
- Since 2020 **HACK Pittsburgh**, *Member and Educator*
- Since 2018 **Software and Societal Systems Department Forge Makerspace**, *Founding Member*, Carnegie Mellon University
- Since 2014 **Collaborative Institutional Training Initiative**, *Responsible Conduct of Research (Physical Science, Social & Behavioral Research)*, Carnegie Mellon University
- Since 2014 **The Corporation of the Seven Wardens**, *Order of the Calling of the Engineer*, Camp 1, Toronto, Ontario, Canada
- Since 2013 **Institute of Electrical and Electronics Engineers (IEEE)**, *Professional Member*, 92797682
Computer Society Technical Communities on Pattern Analysis and Machine Intelligence and Security and Privacy; IEEE Internet of Things, Digital Privacy, and TechEthics Communities.
- Since 2011 **International Red Cross**, *Class C Emergency First Aid, CPR and Defibrillator Certification*